

DEEP LEARNING: ADVANCED MODELS AND METHODS

Knowledge Distillation for Mobile Action Recognition

Compressing a 3D ResNet-50 into an ultra-lightweight MobileNet3D on UCF-101 via logit-based, attention transfer, and cross-frame distillation.

STUDENT

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TRACK

Track 6 - KD for Mobile Action Recognition

INSTITUTION

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PROFESSOR

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REPOSITORY

github.com/danii909/ResNet-to-MobileNet-Action

The Capacity Gap

Goal: Compress a 3D ResNet-50 teacher (pretrained on Kinetics-400) into a MobileNet3D student for on-device video action recognition on UCF-101 (101 classes).

TEACHER

3D ResNet-50

88.82%

TEST ACCURACY

~128 MB

6.21 ms GPU

76.6 ms CPU

ACCURACY GAP

-29.3 pp

SIZE REDUCTION

13.5×

INFERENCE SPEED

1.82×

1.66× CPU (BS=1)

STUDENT

MobileNet3D

59.48%

TEST ACCURACY

~9.5 MB

3.42 ms GPU

46.1 ms CPU

UCF-101 & Group-Aware Split

101
ACTION CLASSES

13,320
VIDEO CLIPS

24f
@ 112×112 PX

Original Split Structure

UCF-101 is officially divided only into Train and Test partitions, **lacking an actual validation set** required for proper model selection.

⚠️ Data Leakage Risk

Correlated clips from the same source video share actor & background. Random splits cause **spatial leakage** and inflated accuracy.

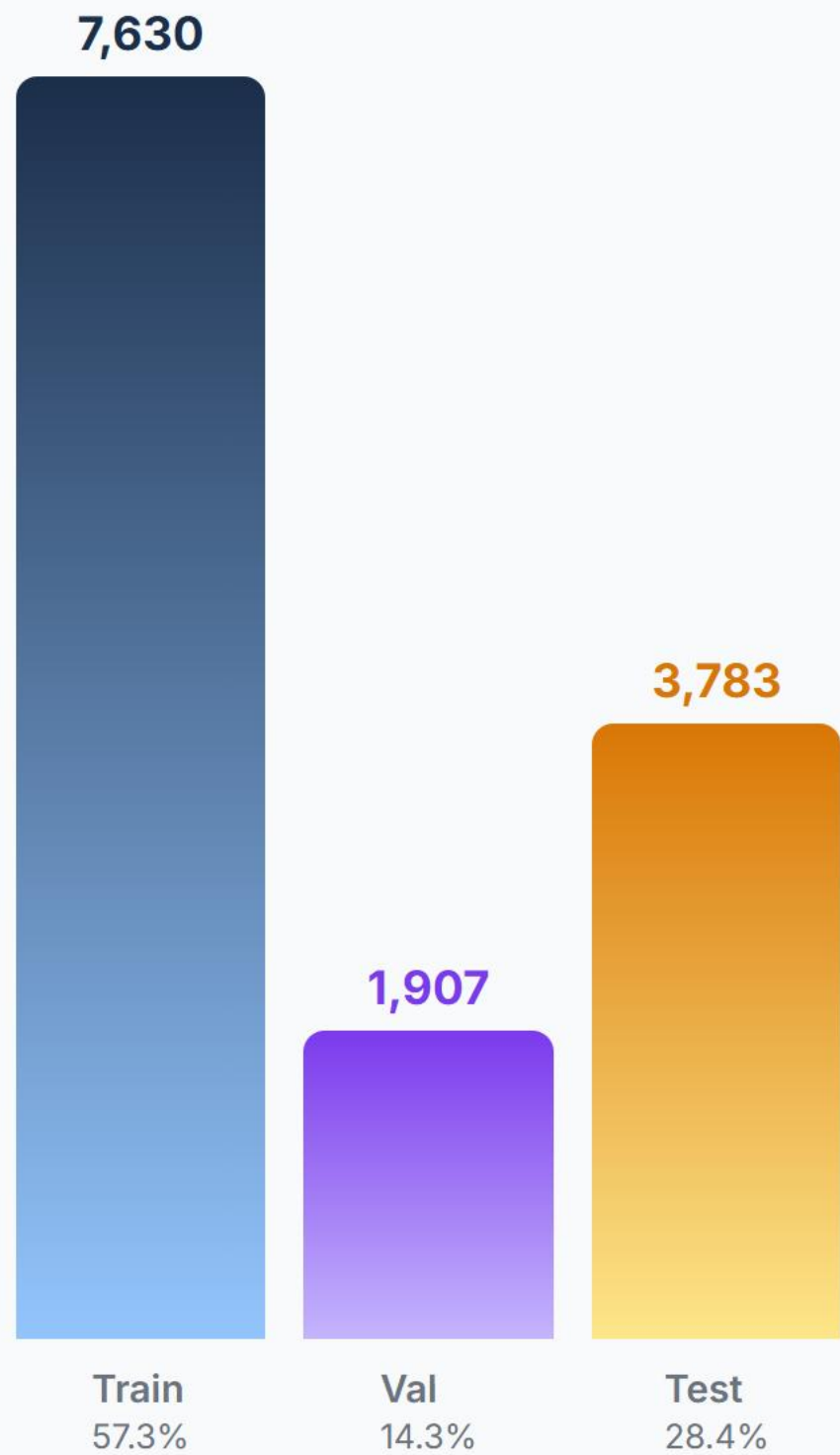
✅ Group-Aware Split

Isolation by group ID (**_gXX_**). No source video shared across Train / Val / Test.

🕒 Temporal Sampling & Stride

Segmented sampling divides the video into N segments, randomly sampling one frame per segment in training (uniform for eval). Stride is dynamic ($\approx L / N$ frames) to span the entire clip.

NEW SPLIT DISTRIBUTION



Knowledge Distillation Loss

Balanced objective function for knowledge transfer

$$\mathcal{L}_{\text{total}} = \underbrace{\alpha \cdot T^2 \cdot D_{\text{KL}} \left(\sigma(z^S / T) \parallel \sigma(z^T / T) \right)}_{\mathcal{L}_{\text{soft}}} + \underbrace{(1 - \alpha) \cdot \mathcal{L}_{\text{CE}}(z^S, y)}_{\mathcal{L}_{\text{hard}}}$$

$\alpha = 0.7$

TARGET BALANCING

Always set to **0.7** for the majority of experiments to prioritize mimicking the Teacher's soft targets (70%) over the hard ground truth labels (30%).

$T \in \{1, 5, 8, 10, 20\}$

LOGIT SCALING

The main hyperparameter varied across experiments to control the smoothness of probability distributions, extracting latent inter-class similarities.

Augmentation Trade-off & Data Leakage

All tests: T=10, a=0.7, 24f · Why light augmentation was chosen

1. Minimal Augmentation

WITH LEAK

65.7%

Top-5: 87.8%

NO LEAK

59.3%

Top-5: 83.6%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Disabled (0.0)
- **Random Erasing:** Disabled
- **Temporal stride:** Fixed (max_stride=1)

2. Light Augmentation ✓

WITH LEAK

69.5%

Top-5: 90.3%

NO LEAK

64.5%

Top-5: 87.4%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Light (0.1)
- **Random Erasing:** 5% probability
- **Temporal stride:** Fixed (max_stride=1)

3. Strong Augmentation

WITH LEAK

55.83%

Top-5: 83.27%

NO LEAK

50.4%

Top-5: 78.5%

CONFIGURATION

- **Spatial crop:** Resized Crop (0.6—1.0)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Strong (0.2)
- **Random Erasing:** 15% probability
- **Temporal stride:** Variable (max_stride=2)

Temperature Sweep & Main Result

Systematic analysis of Student behavior as T varies · a = 0.7 fixed

TEMPERATURE	BEST VAL ACC	EPOCH	TRAIN ACC	TEST TOP-1	TEST TOP-5	Δ VS BASELINE
Minimal Augmentation (no KD)	59.18%	—	99.81%	59.48%	82.77%	ref.
T = 1	62.75%	60	98.44%	62.91%	85.59%	+3.43%
T = 5	63.12%	57	95.87%	62.07%	87.02%	+2.59%
T = 8	64.57%	56	97.59%	64.31%	87.68%	+4.83%
T = 10	64.85%	60	98.28%	64.47%	87.44%	+4.99%
T = 20	65.18%	59	99.05%	65.85%	87.81%	+6.37%

TEST ACCURACY VS. TEMPERATURE TREND



Latent Space Analysis via t-SNE

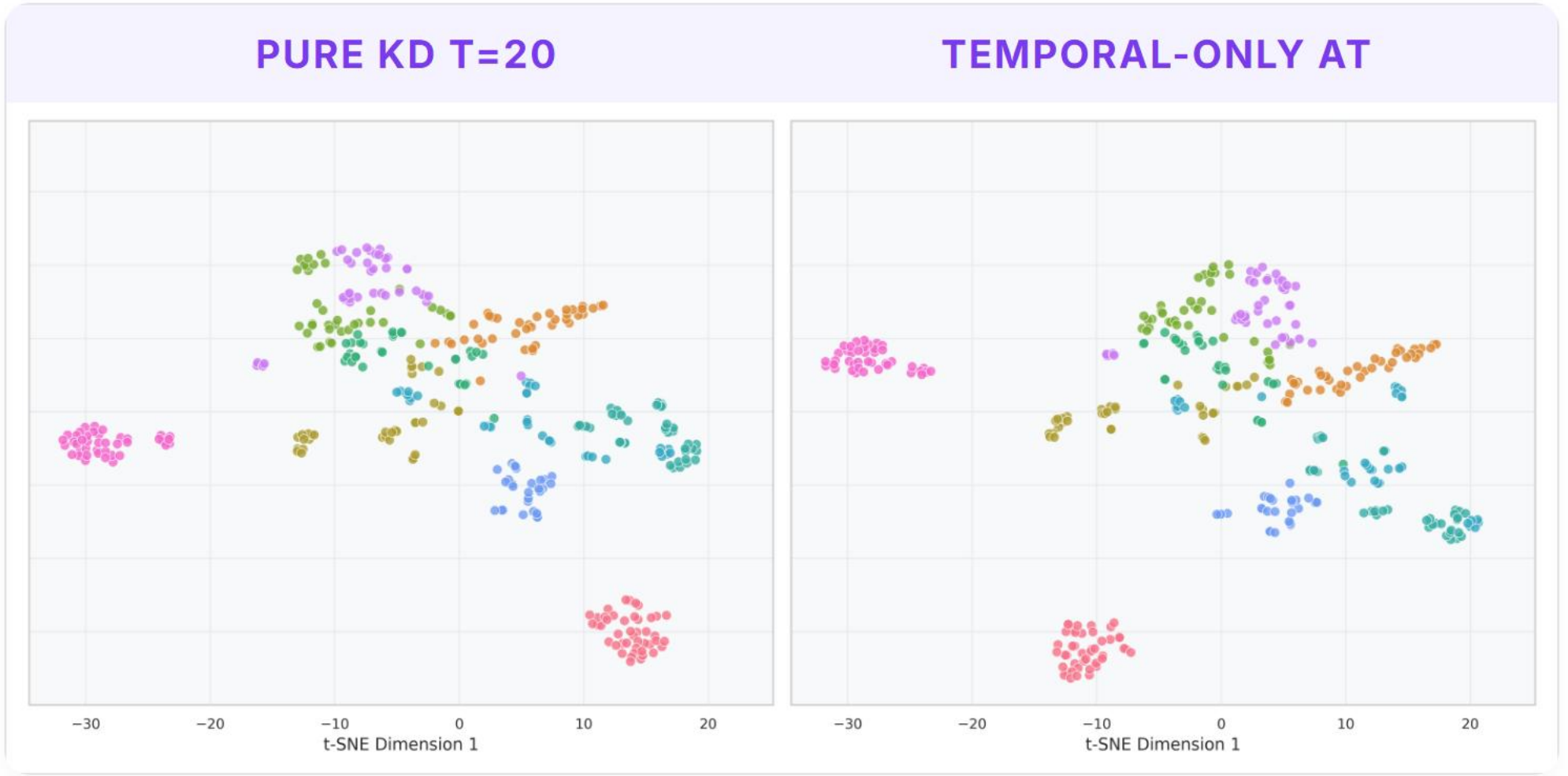
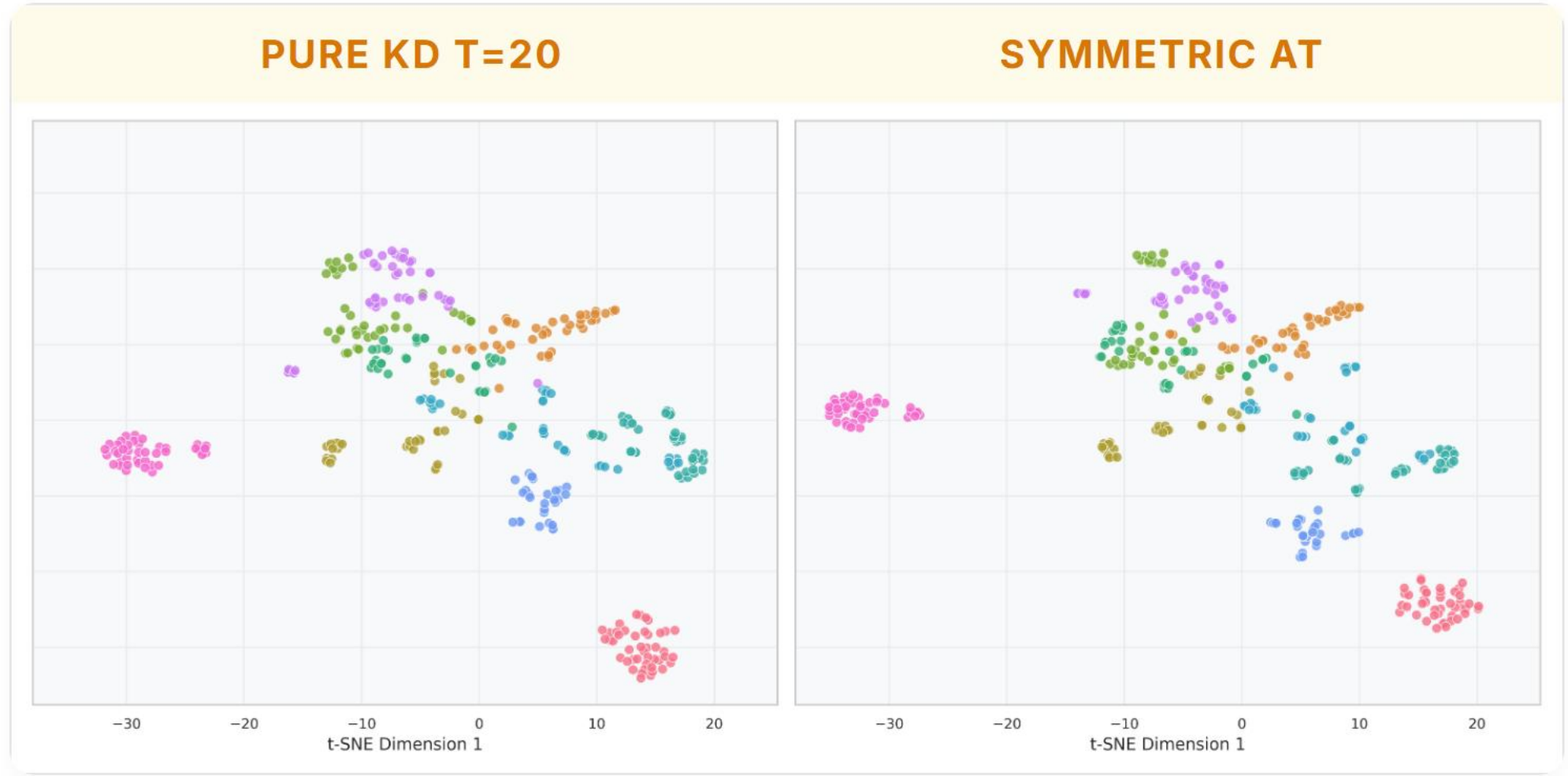
2D projections of pre-classifier embeddings — 10 selected classes



The distilled model (T=20) shows significantly tighter and more distinct class clusters compared to the baseline student, successfully inheriting the well-separated semantic boundaries of the teacher model.

Attention Transfer (AT)

Capacity gap impact on intermediate feature alignment



PURE KD T=20

65.85%

CAPACITY LOSS

-1.27%

→

SYMMETRIC AT

64.58%

Forced spatial alignment acts as noise due to Student's depthwise convolution capacity limit.

PURE KD T=20

65.85%

CAPACITY LOSS

-0.82%

→

TEMPORAL AT

65.03%

Removing spatial constraints ($\beta_s=0$) attenuates regression, as temporal dynamics are more transferable.

Born-Again Networks & Knowledge Collapse

Iterative self-distillation MobileNet3D → MobileNet3D

$\alpha = 0.5, T = 2.5$

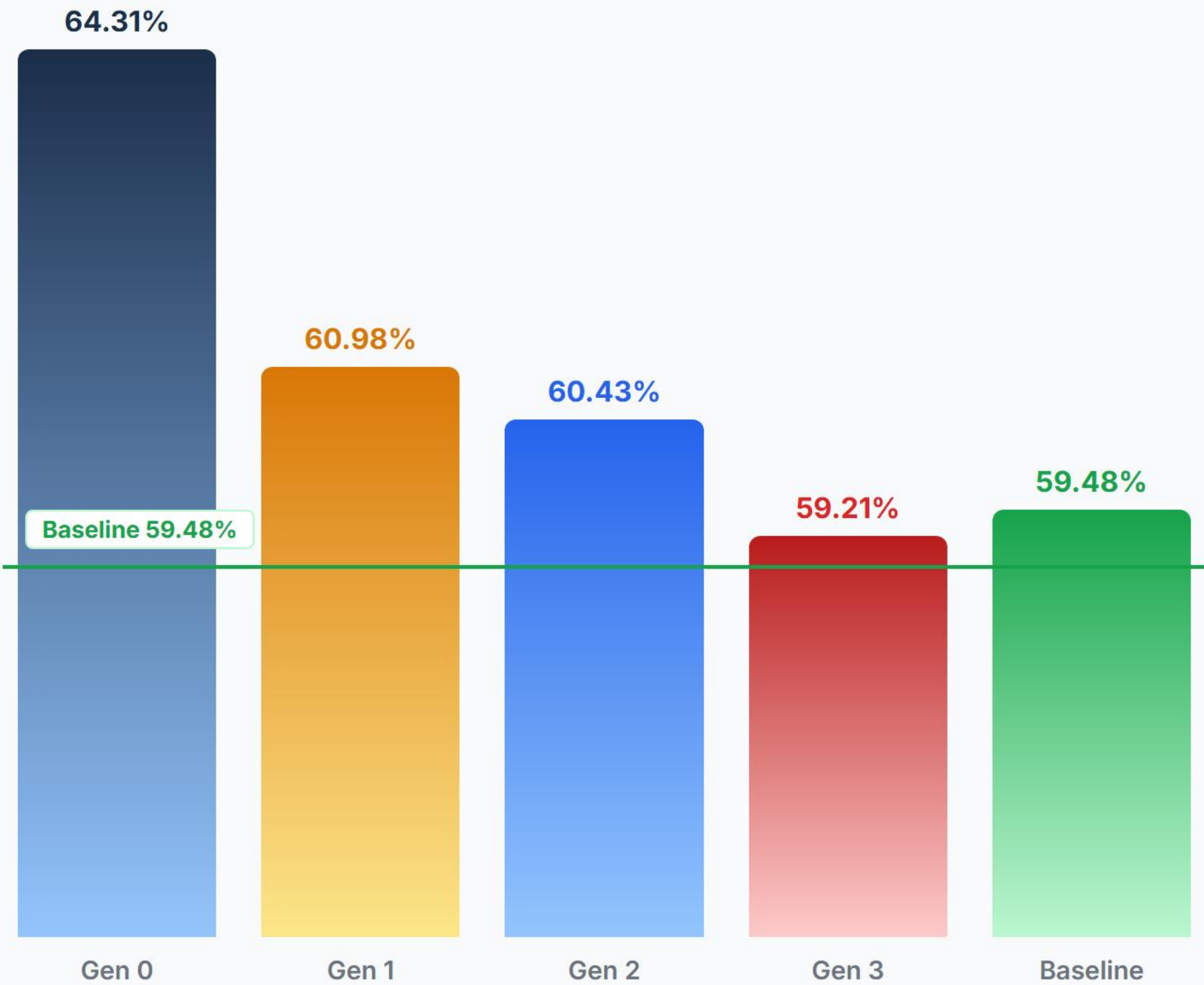
Iterative Self-Distillation

GEN	TEACHER	T	A	ACCURACY
Gen 0	3D ResNet-50	8.0	0.7	64.31%
Gen 1	MobileNet3D Gen 0	2.5	0.5	60.98%
Gen 2	MobileNet3D Gen 1	2.5	0.5	60.43%
Gen 3	MobileNet3D Gen 2	2.5	0.5	59.21%
Baseline	No Teacher (Scratch)	—	—	59.48%

Considerations

A compressed student (64.31%) lacks the visual representational capacity to act as an effective teacher. Distilling at **T=2.5** fails to prevent error propagation, causing progressive degradation back to baseline (59.48%).

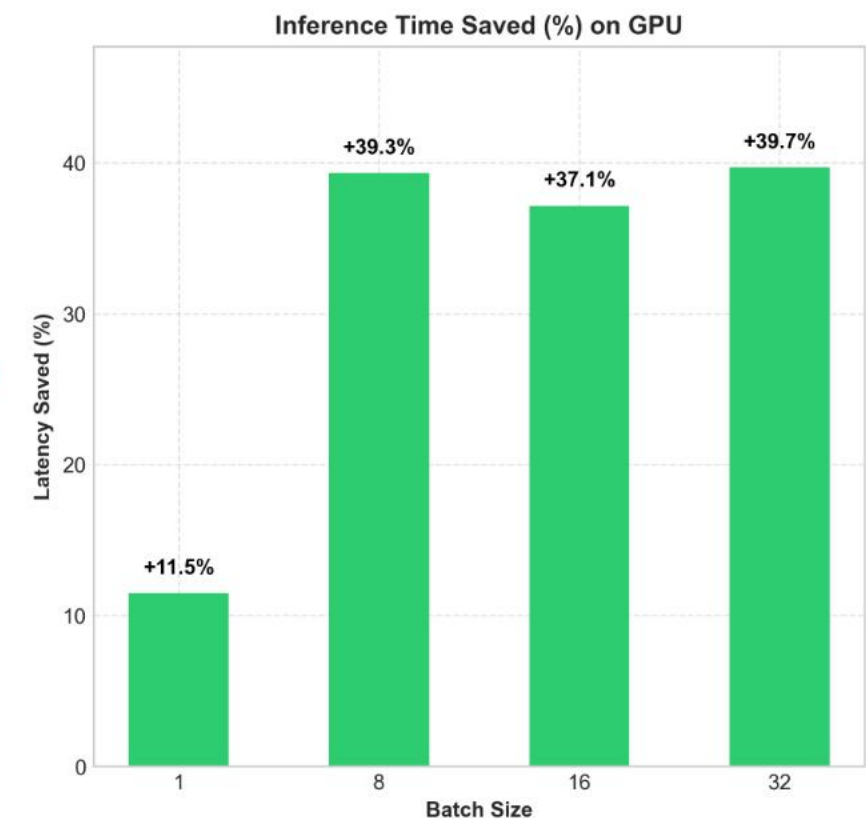
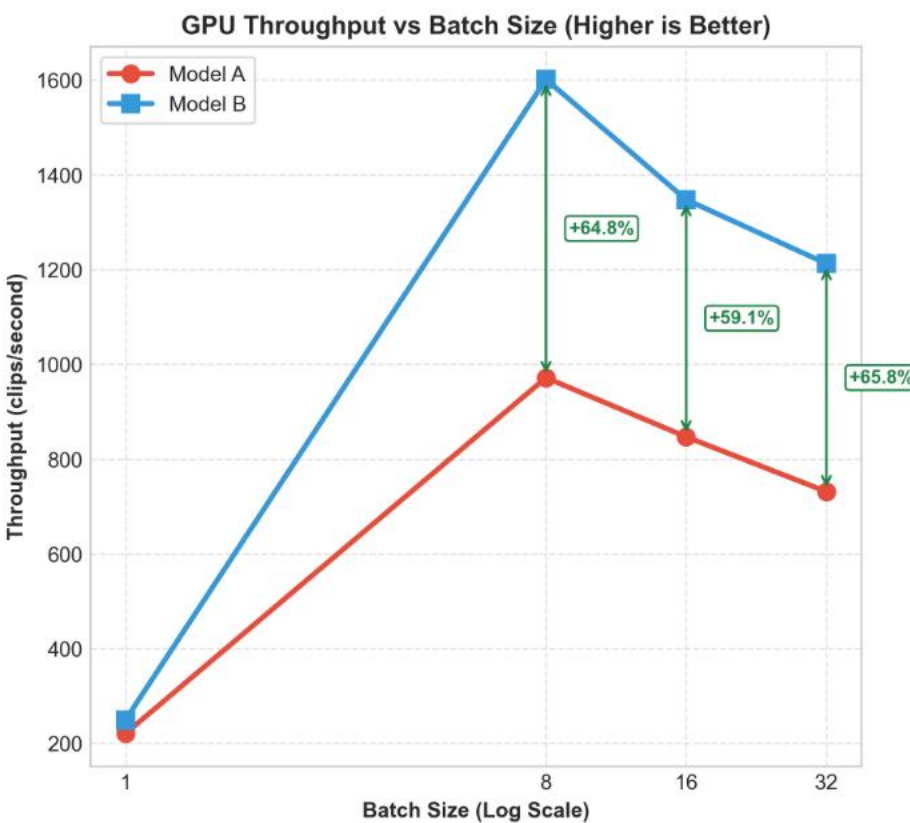
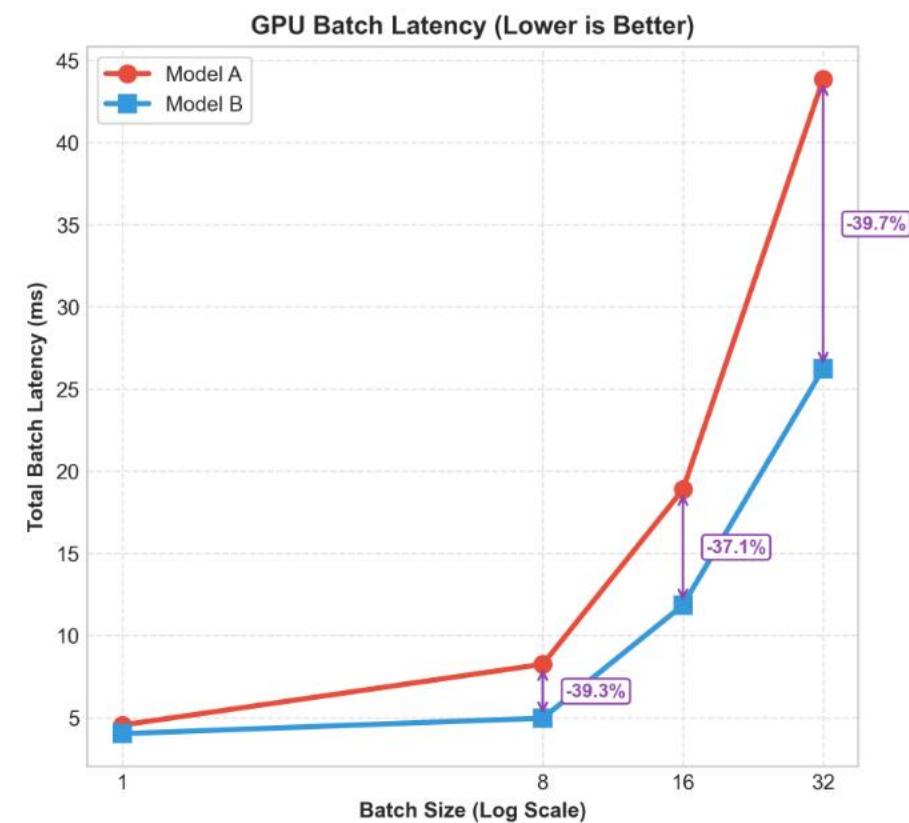
ACCURACY ACROSS GENERATIONS



Cross-Frame Distillation

Asymmetric temporal sub-sampling (T=20, a=0.7) — 16f Student vs 24f Teacher

Test set inference Benchmark (GPU & CPU): Model A (24f) (Top-1: 65.85%) vs Cross Frame Model B (16f) (Top-1: 62.68%)

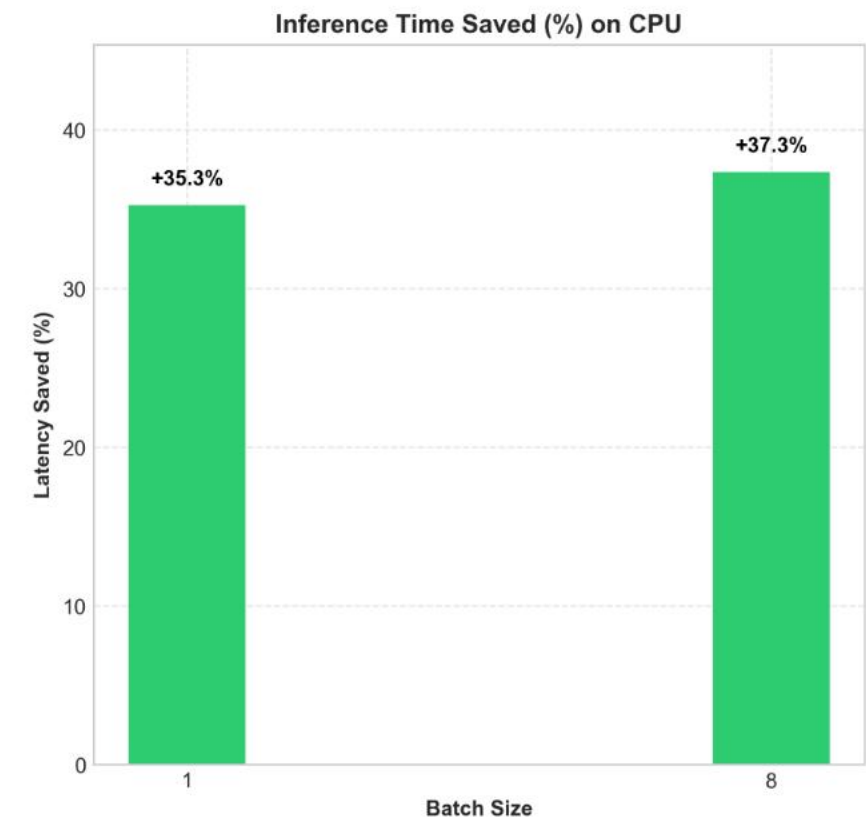
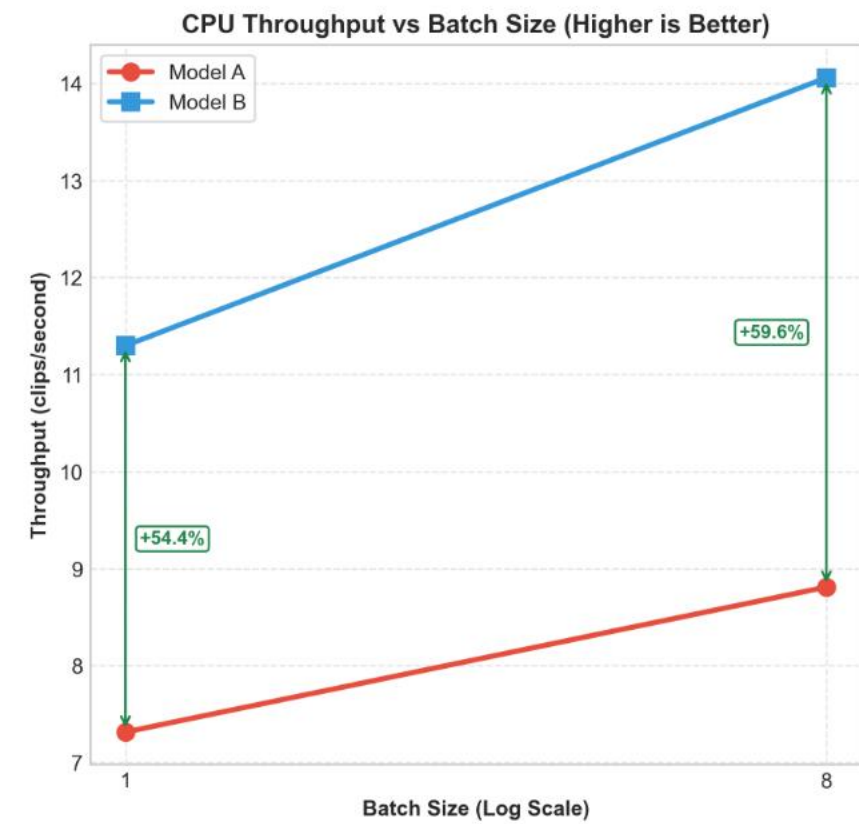
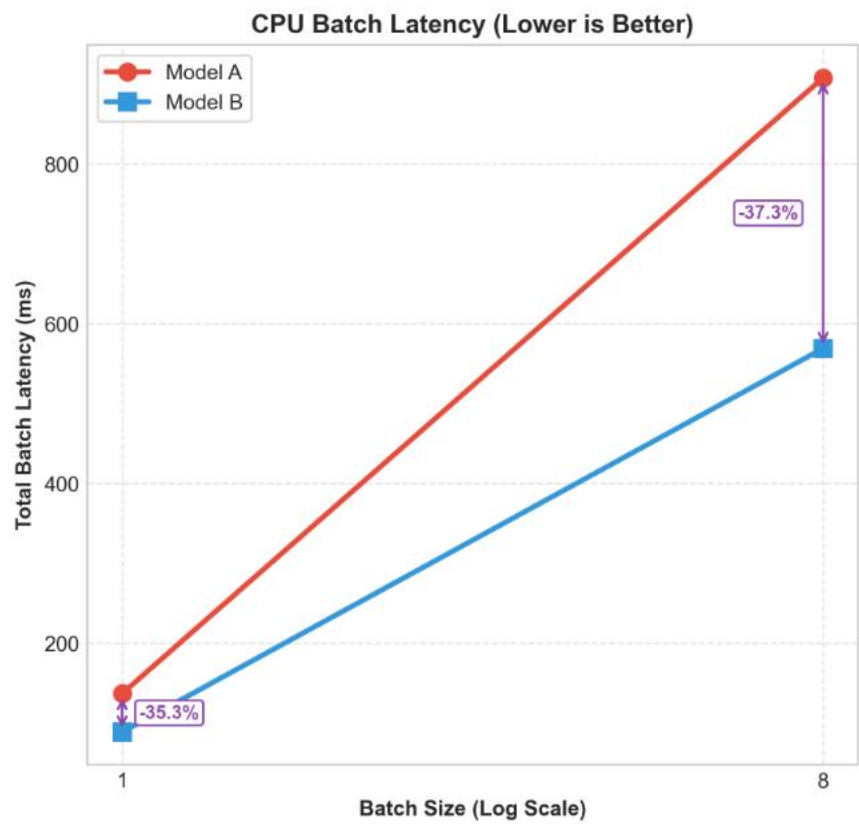


PERFORMANCE GAP

-3.17%

65.85% (24f) → 62.68% (16f)

Minimal accuracy cost for temporal compression



INFERENCE SPEEDUP

2.11×

CPU latency halved (-52.6%)

GPU throughput boosted by +68.8%

Key Takeaways & Future Work

NO FRAME DIST. (24F)

ACCURACY

65.85% +6.37 pp

vs. 59.48% Baseline

CPU LATENCY

53.52 ms

24-frame computational load

GPU THROUGHPUT

991 v/s

Standard throughput

CROSS-FRAME DIST. (16F)

ACCURACY

62.68% +3.20 pp

vs. 59.48% Baseline

CPU LATENCY

25.36 ms -52.6% (2.11x)

Latency halved on Edge CPU

GPU THROUGHPUT

1673 v/s +68.8%

Register & Tensor Core alignment

Lessons Learned

The spatial capacity gap limits intermediate feature transfer. Logit distillation at high temperature (T=20) is the most stable guide.

Key Limitations

- (KD) High temperature (T=20) over-smooths fast, cyclic actions.
- (AT) Spatial capacity gap limits intermediate Feature Distillation.
- (BAN) Error propagation & bias accumulation in Born-Again Networks.

Future Work

- **Variable Temporal Stride:** Train with dynamic frame sampling rates to prevent high-T over-smoothing on fast, cyclic actions.
- **Flexible Feature Alignment:** Introduce temporary learnable adapter blocks (e.g. 1x1x1 convs) during training to bridge the AT capacity gap.

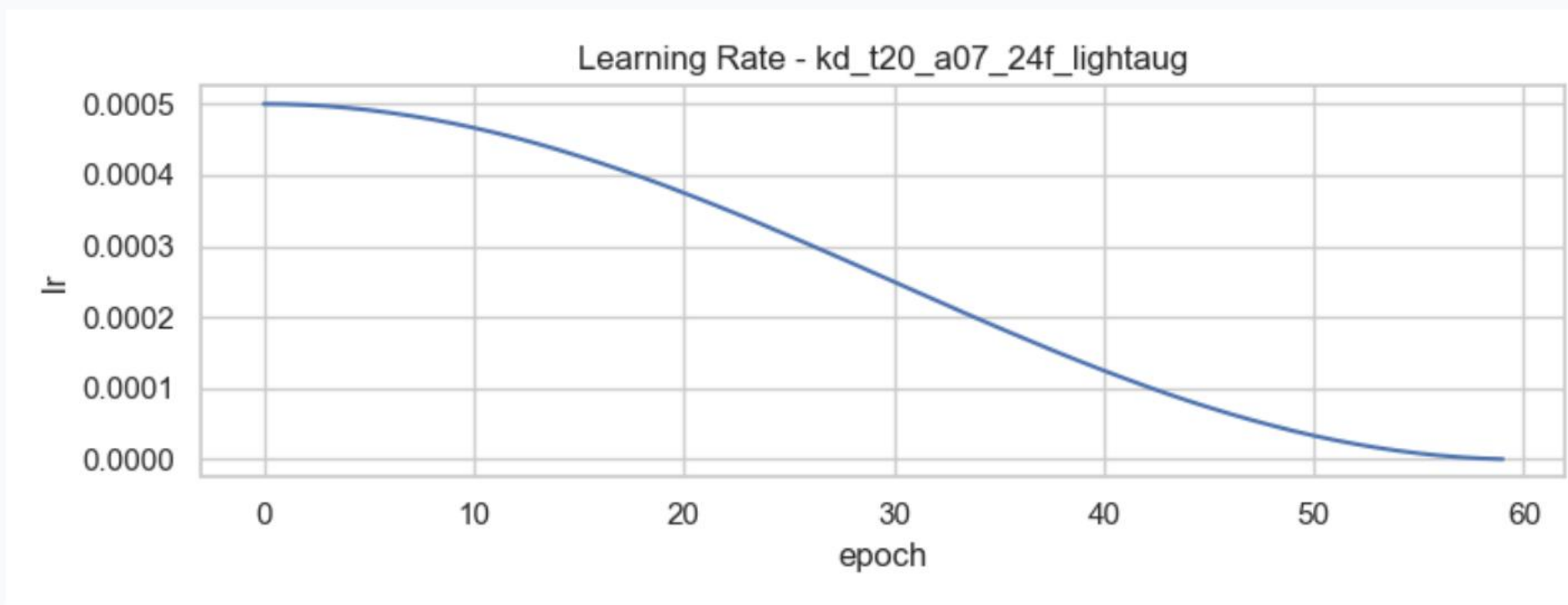
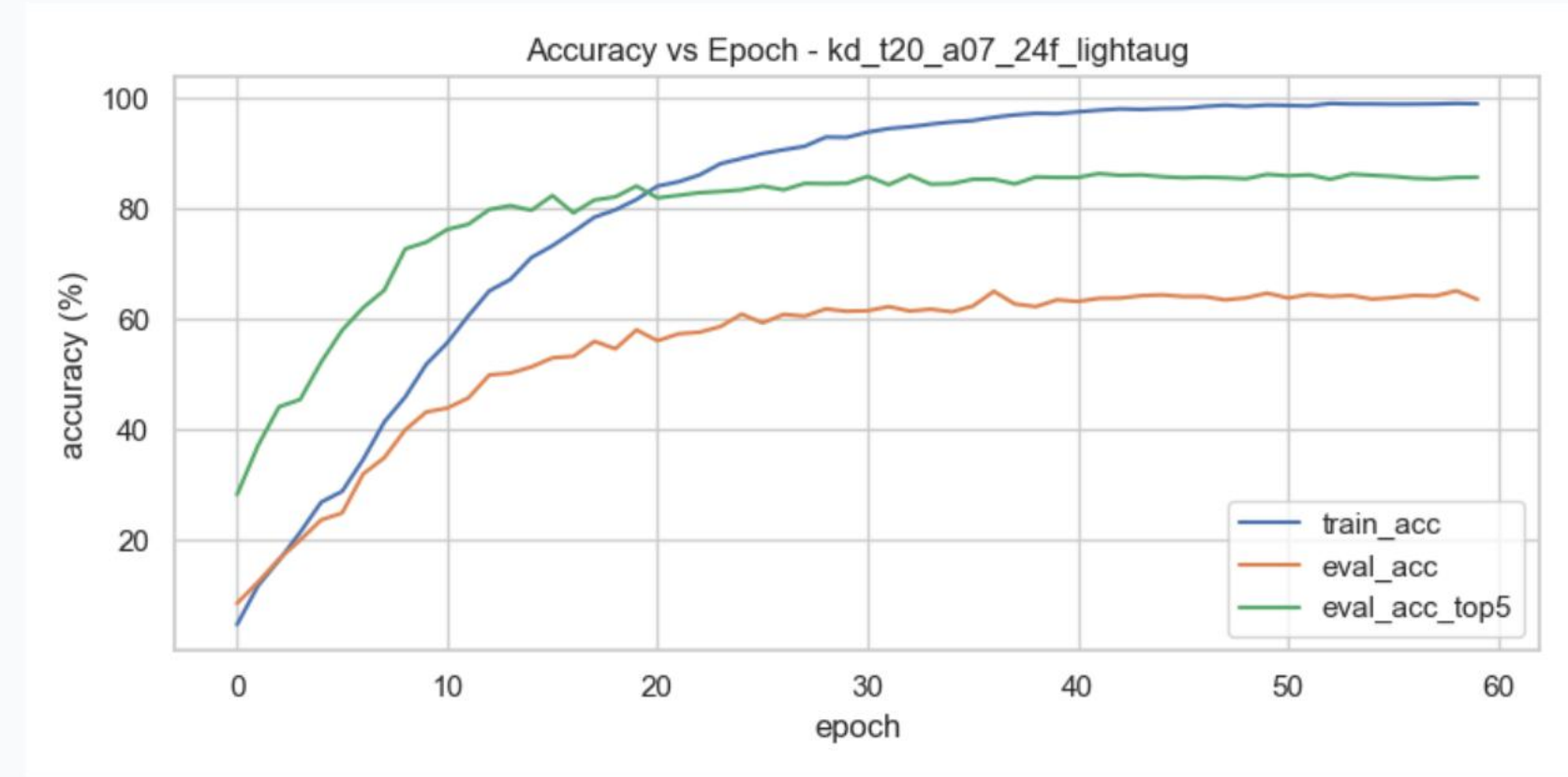
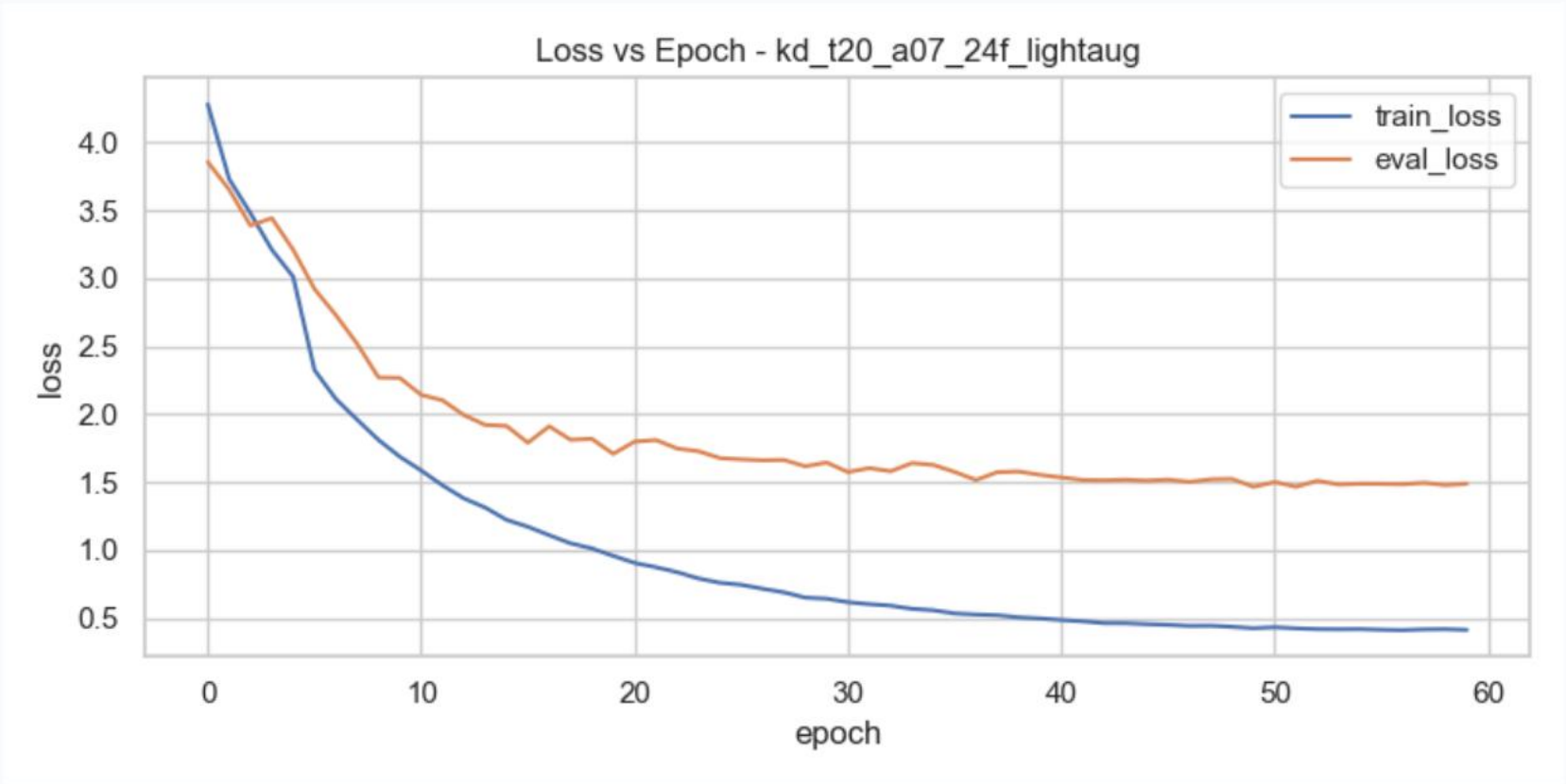
ADDITIONAL MATERIAL

Backup Slides

Quantitative plots, qualitative detailed analyses, and CM predictions.

Training Curves — KD T=20, a=0.7

Loss, accuracy, and LR schedule over 60 epochs



Stability & Convergence

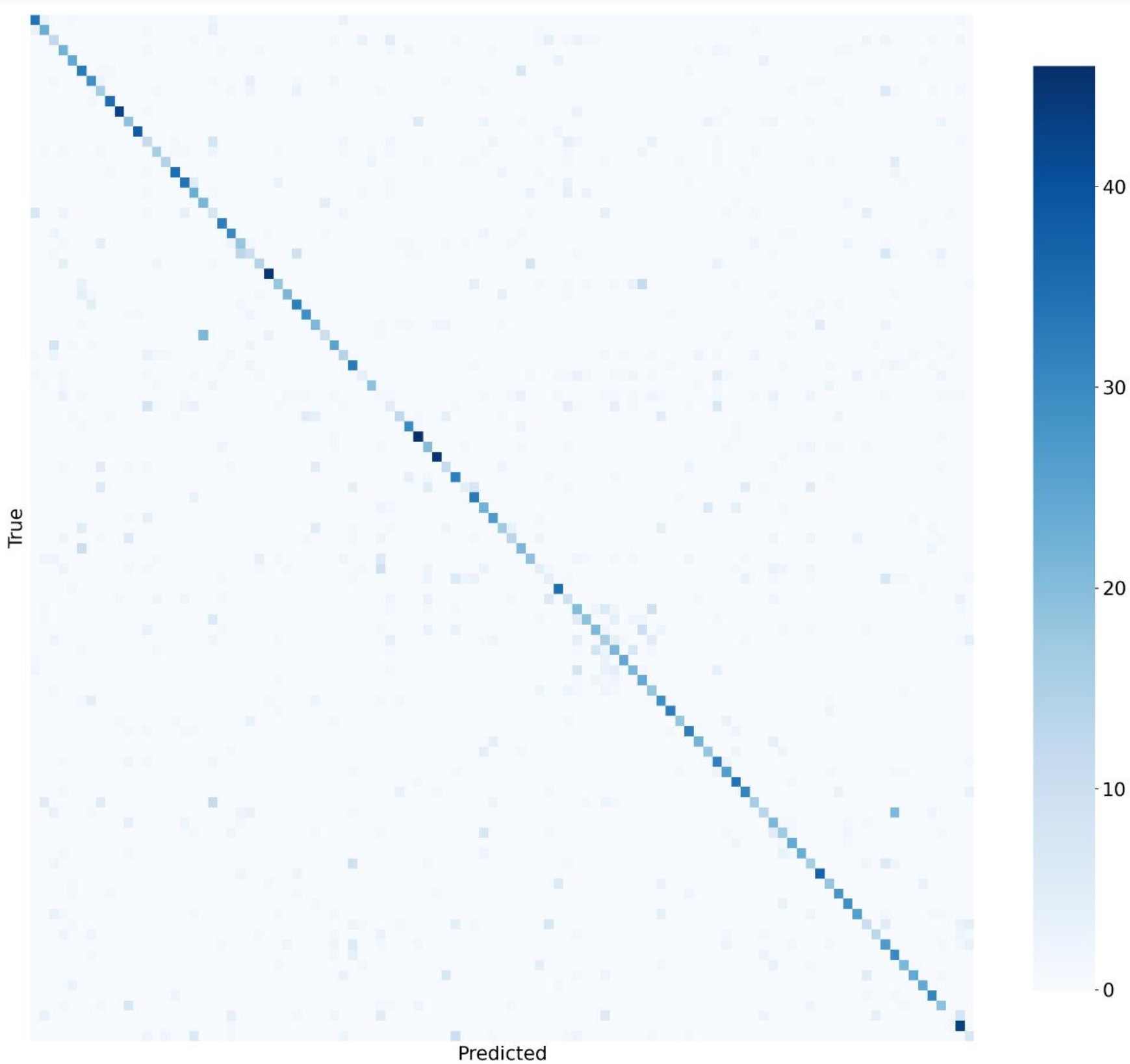
Stable training over 60 epochs. First 10 epochs serve as warmup with hard targets before soft distillation loss activates.

No strong overfitting divergence — regularization works optimally with T=20.

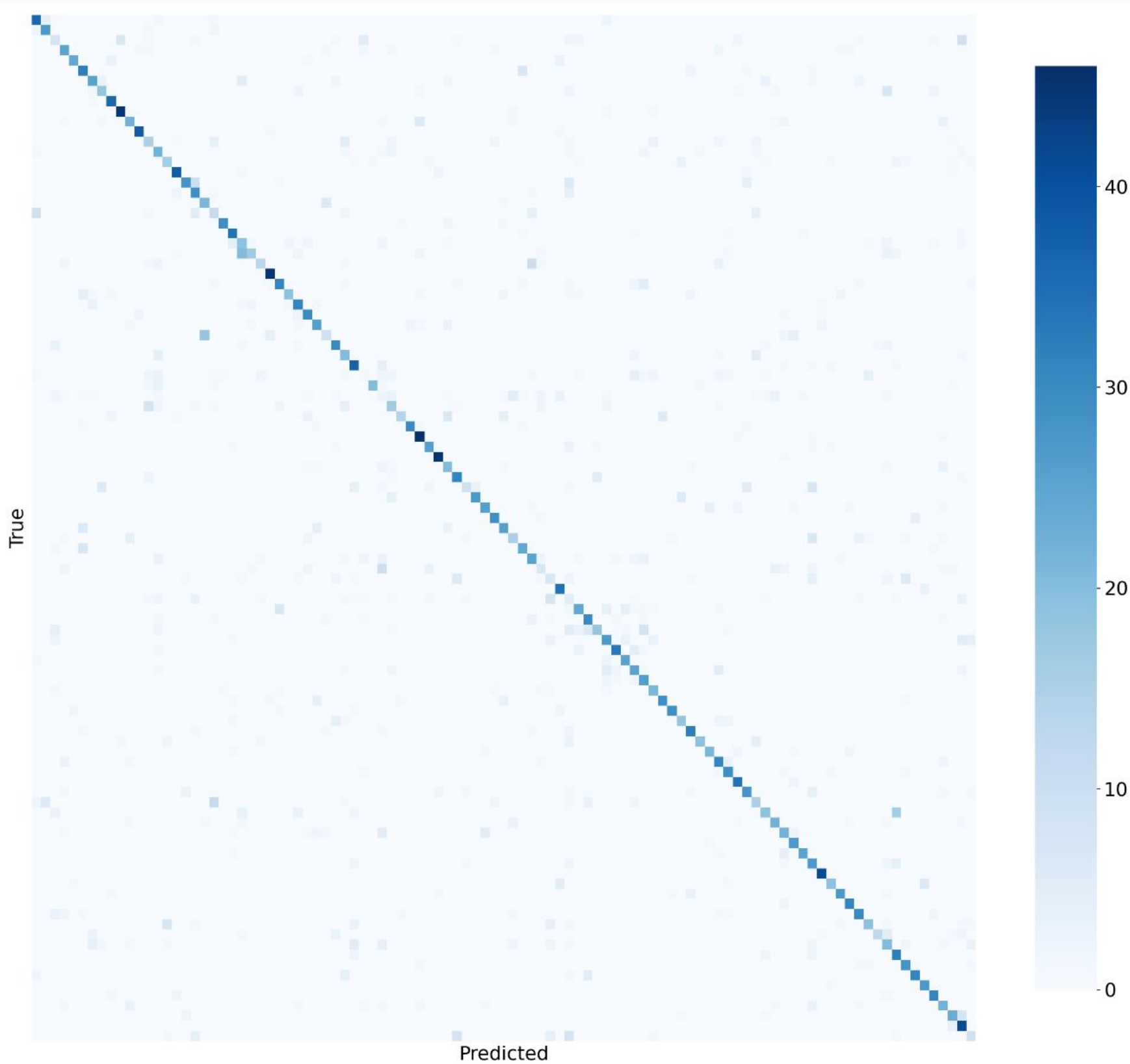
Confusion Matrix & Error Analysis

Baseline vs KD T=20 - Full normalized matrices

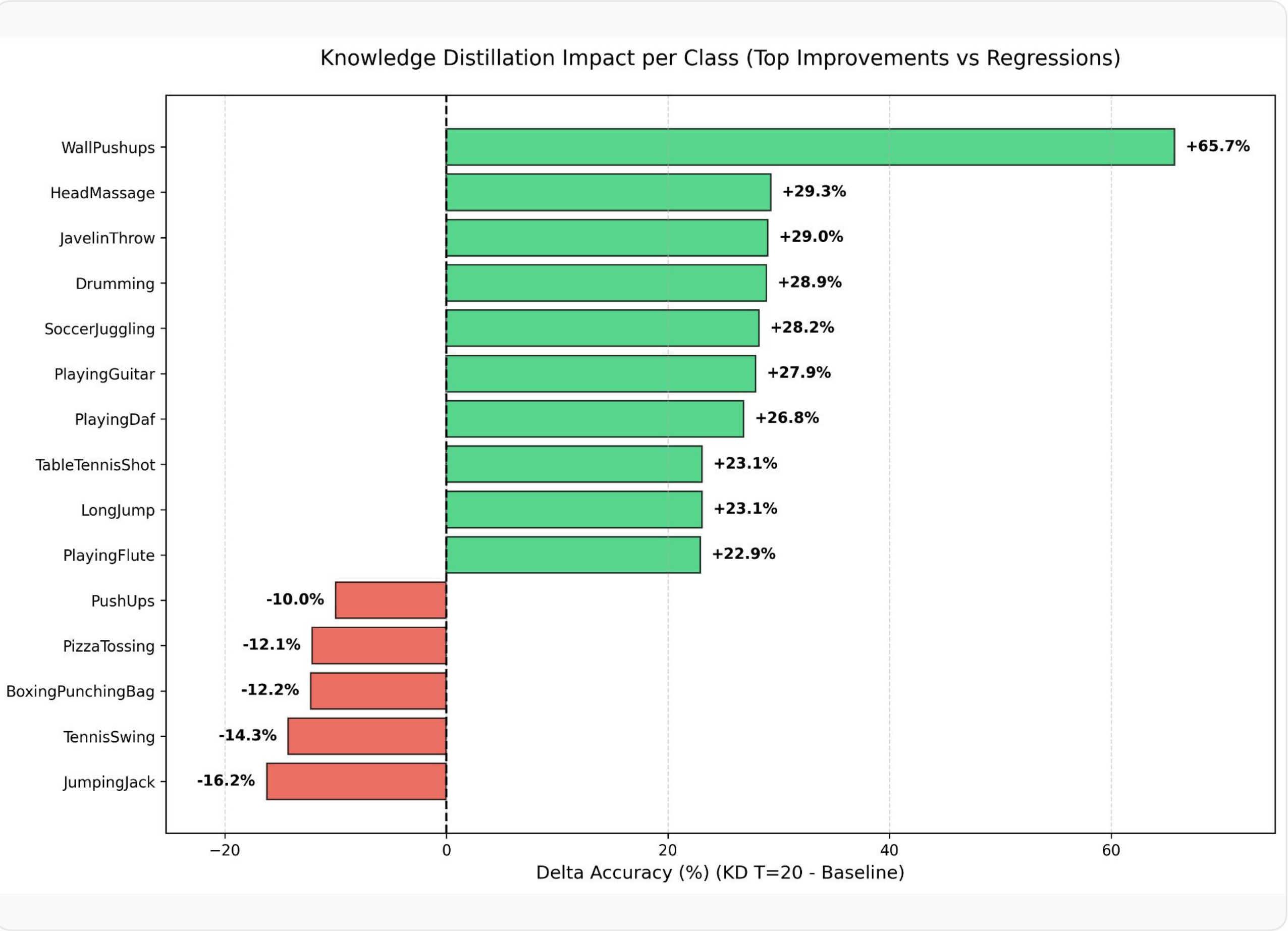
BASELINE - 59.48%



KD T=20 - 65.85%



Per-Class Accuracy: KD T=20 vs Baseline



Improvements

Spatially complex classes (e.g. **WallPushups**, **HeadMassage**) gain most from the Teacher's visual filters.

Regressions

Fast, cyclic actions (**JumpingJack**, **TennisSwing**) suffer from temporal over-smoothing at high T.

Shared Challenges

Classes with high structural ambiguity (e.g. **HandstandWalking**, **Nunchucks**) or strong mutual confusion (e.g. **MoppingFloor**) remain unresolved in both models.

Attention Transfer Loss: Before and After

Decomposing 3D video features to resolve the spatial-temporal collapse

Before: Single Collapsed Loss

$$\mathcal{L}_{AT} = \sum_l || \text{normalize}(\text{mean}_c(F^2))_{flat} - ... ||^2$$

- **Block Mismatch:** Hooked ResNet stages [3, 4, 5]. Stage 5 is the classifier head (produces 2D logits [B, 101]), which caused runtime crashes and zero gradient contribution.
- **1D Interpolation Mismatch:** Resizing a flattened 1D vector of shape [T × H × W] treats space and time as a single coordinate line. This mathematically mixed pixels across frame boundaries and unrelated spatial points, destroying feature geometry.
- **Temporal Signal Dilution:** Normalizing the entire flattened space meant temporal peaks were washed out by spatial dimensions.

59.14%

After: Spatial & Temporal Decomposition

$$\mathcal{L}_{AT} = \beta_s \mathcal{L}_{spatial} + \beta_t \mathcal{L}_{temporal}$$

- **Correct Block Pairing:** Hooked ResNet stages [2, 3, 4] (intermediate 3D features: [B, C, T, H, W]), correctly matching student stages [2, 4, 6].
- **2D Bilinear Interpolation:** Spatial attention is calculated as mean_{c,t}(F²) and resized spatially via 2D bilinear interpolation, preserving frame layout.
- **Temporal AT Isolation:** Temporal attention is calculated as mean_{c,h,w}(F²) yielding [B, T], which is transferred directly without any interpolation noise.

64.58% · 65.03%

Symmetric: β_s=0.05, β_t=0.05 · Temporal: β_s=0, β_t=0.10